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Governing digital platform ecosystems for social options

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Abstract

To date, there is little insight into how digital platforms might be governed towards the creation of social value. We argue that digital platforms can contribute to the creation of social value by enabling *social options*. Thus, we are concerned about how digital platform ecosystems may be governed towards the enablement of social options. We report on the results from our longitudinal qualitative case study of a digital platform employed by ministries of health in more than 70 low- and middle-income countries. We trace the formation and subsequent shaping of governance mechanisms employed to enable social options across national borders and domains. We develop a model of platform governance that shows how such mechanisms are shaped over time through *resourcing*, *capacitating* and *purposing* processes. These three processes play distinct yet complementary roles in governing for social options.

KEYWORDS

digital platform, ecosystem, governance, social options, social value

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1 | INTRODUCTION

Digital platforms[1] have emerged as dominant models for value creation and capture in the global economy (Bosch, 2009; Boudreau, 2010; Evans & Gawer, 2016). By establishing themselves as ecosystem focal points (Jacobides et al., 2018), platform leaders capture economic value together with the distributed actors[2] that complement their digital platforms (Evans et al., 2008; Gawer & Cusumano, 2014). Recently, scholars have called for a deeper understanding of how non-financial forms of value can be derived from digital platform ecosystems (Chamakiotis et al., 2021), for example, by contributing to public service delivery and social development (Bonina et al., 2021; Brown et al., 2017; Gong & Li, 2023). Specifically, information systems (IS) researchers have expressed an interest in how digital platform ecosystems can generate *social value*. Social value constitute "a non-economic type of value exceeding individual interests and impacting on a wider community" (Chamakiotis et al., 2021, p. 95), such as by enabling inclusive public service delivery and consumption (Vassilakopoulou et al., 2017) or new forms of knowledge generation and organising (Ameen et al., 2024; Chamakiotis et al., 2021).

In this article, we examine how digital platform ecosystems can be governed towards the enablement of social options (hereafter, governed *for* social options)—*common, open, and generic resources that afford a wide range of possibilities for social value creation* (Frischmann, 2012). Social options are shared and accessible resources for distributed actors to experiment on, learn and adapt in their efforts to create social value (Benkler, 2013; Frischmann, 2004, 2012). Maintaining the common, generic, and open nature of resources 'precludes optimization for a narrower range of activities' (Frischmann, 2012, p. 94) and grant actors autonomy in value creation. Extending Frischmann's (2012) notion of social options to governance in digital platform ecosystems, we add that governing for social options involves *favouring activities that are regarded as beneficial to a community over those that are primarily advantageous to individual actors*.

Governance of digital platforms requires balancing the welfare of the ecosystem with the autonomy of its heterogeneous actors (Parker & Van Alstyne, 2017). Governing for social options presents a distinct version of this general dilemma. On one hand, governance should defer decisions on how to leverage common digital platform resources to complete the platform's design and value propositions (Garud et al., 2008). Hence, governing for social options entails enabling individual actors to generate social value that is contingent on local needs and contexts. Social options may be leveraged in unanticipated ways and produce rich spillover effects in terms of innovation, knowledge, and other social goods. This generative potential resonates with well-known value co-creation dynamics in digital platform ecosystems (Constantinides et al., 2018; de Reuver et al., 2018). On the other hand, a too open governance regime diminishes the platform leader's ability to *orchestrate the ecosystem actors' activities towards communal goals and values* (Cennamo & Santaló, 2019; Karhu et al., 2018). Ecosystem actors' distributed creation of knowledge and innovations must be governed in a way that enables further social options for the ecosystem. Platform leaders who seek to govern for social options must thus facilitate the distributed autonomy of heterogeneous actors *and* simultaneously foster collective action.

This dilemma ties governance of digital platforms for social options to an IS research tradition of seeking alternative governance mechanisms to address collective action challenges. As Monge et al. (1998, p. 413) early on pointed out, widespread participation is essential in 'joint value creation alliances' because different actors make qualitatively different contributions. As digital platforms attract heterogeneous contributing actors, aligning their interests and activities becomes a governance challenge (Cennamo & Santaló, 2019). The variety of applied governance tactics should therefore address collective action challenges (Markus et al., 2006). According to Ostrom's (1990) observation, collective management of common resources (in our case, social options) depends on factors internal to the

[1]In this paper, we refer to digital platforms or software platforms that serve as bases for innovation (innovation platforms), as opposed to digital platforms that primarily function as marketplaces (transaction platforms).

[2]We use the term *actor*, rather than complementor, application developer or partner, to recognise the heterogeneous and multifaceted roles of contributors in digital platform ecosystems.

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community, such as communication and trust. In turn, these internal characteristics relate to the values and ideologies of the actors involved and the legitimation of collective interests (Constantinides & Barrett, 2015).

We approach the dilemma of governing digital platform ecosystems for social options as a problem that requires navigation over time and pose the research question: *How can governance mechanisms be shaped to enable social options in a digital platform ecosystem*? We take a processual perspective on digital platform ecosystem governance and emphasise that governance mechanisms are socio-technical, and they evolve over time. Platform leaders introduce and modify governance mechanisms, while other ecosystem actors push against their constraints in the pursuit of specific goals (Eaton et al., 2015; Ghazawneh & Henfridsson, 2013; Huber et al., 2017). Previous research has identified how governance mechanisms, which define how ecosystem actors interact with and extend the platform, provide the means to regulate the actors' actions and outputs (Eaton et al., 2015; Gregory et al., 2018). Our study extends these insights by investigating how the shaping of governance mechanisms unfolds when enabling social options serves as a guiding principle for digital platform governance.

To answer our research question, we study the governance of a digital platform used for public health management in low- and middle-income countries. Data from participant observation, meeting minutes, email exchanges, interviews, and naturally generated documentation from the period 2015–2022 inform our engaged and longitudinal qualitative case study of the web-based and open-source District Health Information Software version 2 (DHIS2). The Health Information Systems Programme (HISP) coordinates DHIS2 development and implementation (Braa et al., 2007; Sanner et al., 2014). The HISP Centre at the University of Oslo (HISP UiO)—the platform leader – acts as the coordinating nexus for software development, capacity building, and knowledge and innovation sharing in a global network of academic institutions, ministries of health, non-governmental organisations (NGOs) and funding agencies (Braa et al., 2004; Roland et al., 2017; Titlestad et al., 2009).

We contribute to the literature on digital platform governance by developing a model of how resourcing, capacitating and purposing processes shape governance mechanisms towards enabling social options. These governance processes are informed by limitations in the realisation and innovation of social options in the form of unmet needs, constrained participation and misalignment between individual actors' activities and communal values. *Resourcing* opens and broadens the scope and diversity of the digital platform. *Capacitating* allows ecosystem actors to participate in the realisation and innovation of social options. *Purposing* favours the actors' activities and outputs that are aligned with communal goals and values. Our findings highlight the continuous shaping of governance mechanisms through these three processes to bring about the realisation and innovation of social options.

In Section 2, we review the literature on digital platform ecosystem governance as a basis for our examination of how platform governance for social options is carried out. In Section 3, we present the background for our empirical case and our engaged insider approach to data collection and analysis. Six empirical episodes that describe resourcing, capacitating and purposing governance processes follow in Section 4. In Section 5, which covers the discussion and contributions of this study, we theorise governing digital platforms for social options and derive a process model of how governance mechanisms are shaped over time. In the final section, we provide our overall reflections and outline this study's limitations and further research directions.

2 | CONCEPTUAL BACKGROUND

By governance mechanisms, we refer to patterns of socio-technical activities and structures that (a) shape who decides what and (b) ensure desirable activities and outputs (Gregory et al., 2018; Tiwana, 2013). Key structural governance mechanisms in digital platform ecosystems are standardised interfaces, automated processes, and multilayered architectural arrangements (Bonina & Eaton, 2020; Gregory et al., 2018). These are technical means to shape social dynamics, enabling a variety of actors to create and derive value from the same digital platform. Governance mechanisms are not established once and kept stable but evolve from the orchestration of a platform leader (Tiwana, 2013) and the interactions among heterogeneous actors (Eaton et al., 2015). To establish a conceptual

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background for our investigation, we first relate digital platform governance to social options before honing in on cooperation and collective action problems in digital platform ecosystems.

2.1 | Digital platform governance and social options

The platform model requires platform leaders to "focus less on managing product and service development within their boundaries, focusing instead on the careful governance of [ecosystem actors]" (Foerderer et al., 2019, p. 120). Platform leaders provide Application Programming Interfaces (APIs), Software Development Kits (SDKs), code libraries, software licences and documentation as key forms of governance mechanisms to steer platform development in a desirable direction. These standardised interfaces enable interactions between the platform and its complementing actors (Dal Bianco et al., 2014; Ghazawneh & Henfridsson, 2013). Ghazawneh and Henfridsson (2013, p. 174) coined the term *boundary resources* to refer to "software tools and regulations that serve as the interface for the arm's length relationship between the platform owner and the application developer". The authors further describe how the platform leader initiates and shapes boundary resources through *resourcing* and *securing* processes. Resourcing refers to processes that open the platform to broaden its scope and diversity, whereas *securing* increases control over the digital platform and its complements.

Longitudinal studies of complex digital platform ecosystems, such as enterprise software platforms (Schreieck et al., 2021) and digital platforms in healthcare (Fürstenau et al., 2019), highlight the need for platform governance that go beyond structural boundary resources (Foerderer et al., 2019; Wareham et al., 2014). These studies show how value co-creation is dynamic and knowledge-intensive and involves extensive cooperation (Bergvall-Kåreborn & Howcroft, 2014; Foerderer et al., 2019; Huber et al., 2017; Sarker et al., 2012). Platform leaders thus draw on additional governance mechanisms, including training, partnership programmes and direct support, to furnish ecosystem actors with knowledge and tools to co-create and integrate platform resources (Bergvall-Kåreborn & Howcroft, 2014; Foerderer et al., 2019). Wareham et al. (2014) have found that platform leaders employ partnership hierarchies to differentiate among contributing actors. With this governance mechanism, actors with a strong ecosystem identification and a high value-generation potential move up the hierarchy and gain access to exclusive knowledge-intensive platform resources.

The platform literature has identified balancing between individual actors' self-interests and the ecosystem's welfare as a key governance dilemma (e.g., Parker & Van Alstyne, 2017). Too much delegation of decision rights to the autonomous actors in the ecosystem can hamper sharing, lead to undesirable duplication of efforts and inhibit pooling of resources (Saadatmand et al., 2019). Governance responses to such cooperation problems include financial instruments, such as pricing and revenue-sharing models or legal and intellectual property agreements (Oh et al., 2015; Schreieck et al., 2021; Tiwana, 2013; Wareham et al., 2014). Specifically, Parker and Van Alstyne (2017) consider the platform leader's appropriation of complementary innovations to maximise ecosystem profitability as a possible governance response. Absorbing an innovation and making it available as a shared platform resource reduce its economic value to the actor who created it, while providing spillover effects and resource integration to the entire ecosystem.

Governing for social options presents a distinct version of the governance dilemma outlined above. As illustrated in Figure 1, governing for social options stipulates that the platform's open resources allow individual actors to realise social value locally and that the distributed creation of knowledge and innovations creates further social options

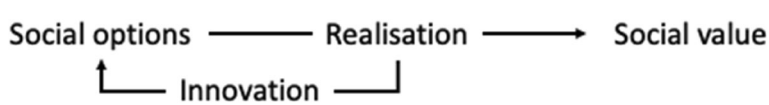

FIGURE 1 Value realisation feeds back to the platform's resources to enable further social options.

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for the entire ecosystem. We use the term *innovation* to signify processes whereby local value realisation feeds back to the platform's shared, open, and generic resources. When social development is an explicit goal (Bonina et al., 2021), the platform leader must ensure individual actors' autonomy for the local realisation of social value, while fostering cooperation in downstream generation of social options. To understand how governance can navigate this dilemma, we turn to the literature on cooperation in digital platform ecosystems.

2.2 | Governance, cooperation, and collective action

Tiwana (2013) alludes to relational control as a governance mechanism that promotes cooperation and sharing in digital platform ecosystems. However, relational control hinges on shared norms and values—'a sort of shared identity that defines the character of the platform ecosystem and rallies [actors] around it by harmonising their own goals with those of the platform' (Tiwana, 2013, p. 125). A platform leader can foster relational control by providing socialisation and training opportunities that espouse shared norms and values or by screening ecosystem entrants. For example, low control pertaining to platform complements can be counterbalanced through strict control over whom to involve in value co-creation activities (Mukhopadhyay et al., 2019). The existing literature provides rich insights into the cooperation dynamics between the platform leader and contributing actors (e.g., Foerderer et al., 2019; Wareham et al., 2014). However, in their recent literature review, Halckenhaeusser et al. (2020) have found that the digital platform literature offers little insight about cooperation among the contributing actors themselves.

The need for cooperation and sharing of resources among contributing actors is particularly pressing in digital platforms established by governments and non-profit organisations since they often adopt digital platforms to enable collaborative service innovation across previously siloed structures (Bygstad & Hanseth, 2018; Cordella & Paletti, 2019; Gong & Li, 2023; Vassilakopoulou et al., 2017). Drawing on insights from three Scandinavian countries, Vassilakopoulou et al. (2017) note that in the public sector context, digital platforms constitute a new form of public good, which foregrounds public service delivery over the realisation of network effects. Given this emphasis, public sector platform leaders face challenges in applying existing platform theory and concepts (Bonina & Eaton, 2020; Brown et al., 2017; Gong & Li, 2023). The ecosystems in question call for alternative modes of value co-creation (Cordella & Paletti, 2019). Digital platform ecosystems developed in the pursuit of non-economic forms of value, such as equitable healthcare, education and social welfare, are undertheorised (Bonina et al., 2021; Koskinen et al., 2018; Sanner & Nielsen, 2019), and empirical accounts of their formation remain few (for exceptions, see, e.g., Islind et al., 2019; Vassilakopoulou et al., 2017).

Individual ecosystem actors may pursue their interests over common goals, which, over time, can depreciate the social options available to all actors. Collective action scholars have long been occupied with similar challenges. Ostrom (1990) has noted alternative institutions where communities self-organise for sustainable management of commons. Communication, transparency, and trust within the groups can substitute for centralised control or privatisation. However, the relatively small-scale natural resource commons examined are often the subjects of a homogeneous group's exploitation. In large-scale joint endeavours, such as a digital platform ecosystem, heterogeneity is desirable, while presenting more complex webs of interests (Markus et al., 2006; Monge et al., 1998). A key learning is that heterogeneity poses collective action problems (Cennamo & Santaló, 2019). Addressing such problems requires the application of multiple combined tactics (Markus et al., 2006), which are subject to changes over time. In the next section, we outline a digital platform governance perspective that is sensitive to this challenge.

2.3 | Governing for social options

Summarising the two preceding sections, governing for social options is a difficult endeavour. It entails a normative direction towards the creation of social value. This directs the capacity around the digital platform towards the needs

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of communities, as opposed to individual actors with the greatest potential to boost network effects and profits. The centralised control inherent in the digital platform model allows the platform leader to favour certain actions and outputs over others based on communal goals and values. At the same time, a restrictive set of governance mechanisms may stifle innovation and value co-creation. We adopt a processual view to discern how this dilemma might be navigated to enable social options.

Governance mechanisms shape how different stakeholders in the ecosystem behave and interact (Karhu et al., 2018). Conversely, governance mechanisms are themselves formed in response to the actions and interactions of ecosystem actors (Eaton et al., 2015; Leong et al., 2024. As shown by Eaton et al. (2015) in their analysis of the iOS ecosystem, 'jail-breaking' emerged as an activity to circumvent rigid control, in turn leading Apple to revise its governance mechanisms. Dynamics such as these directed the entire ecosystem to a more open and generative configuration. Ecosystem actors' objectives and preferences also change throughout their engagement with the platform (Leong et al., 2024, and they serve user needs that evolve and transform (van der Geest & van Angeren, 2023).

Focusing on governance *processes* highlights the longitudinal evolutionary dynamics that shape governance mechanisms over time (Eaton et al., 2015; Huber et al., 2017; Karhu et al., 2018; Leong et al., 2024. It brings out events whereby governance mechanisms "evolve and collide with artifacts within and across multiple organisational and technological contexts" (Eaton et al., 2015, p. 221). Existing research has focused on how socio-technical arrangements inhibit goal-seeking actions of ecosystem actors and how the platform leader might respond by creating and modifying governance mechanisms (Ghazawneh & Henfridsson, 2013; van der Geest & van Angeren, 2023). In the remainder of this article, we empirically discern how governance mechanisms are shaped to align ecosystem activities and outcomes with the guiding principle of enabling social options.

3 | RESEARCH DESIGN

Our case study is based on our long-term engagement with the free and open source DHIS2. The software has its roots in the 1990s as an outcome of an action research project in the Western Cape in post-apartheid South Africa (Braa & Hedberg, 2002). Supporting data-driven and decentralised public sector management has remained a key aspiration of the DHIS2 venture. The HISP UiO coordinates DHIS2 development and implementation. A long history of participatory design, capacity building and the focus on local ownership has contributed to the cultivation of an extensive international research and service network (Braa et al., 2004; Braa & Hedberg, 2002; Roland et al., 2017).

Our research is executed as a longitudinal interpretive case study (Klein & Myers, 1999; Walsham, 1995, 2006). Longitudinal studies are needed to theorise governance of digital platform ecosystems across heterogeneous organisations and settings (Tilson et al., 2010) since such processes are inherently distributed and stretch out over time. As engaged insiders, we have examined the conditions for and the unfolding of governance processes in a complex multinational setting. Empirically, we follow the transition towards DHIS2 as a digital platform, with primary data collected from 2015 to 2022. In the next section, we present the context and background of our empirical case, followed by accounts of our qualitative data collection and analysis.

3.1 | Case context and background

Parting with the standalone DHIS1, the HISP UiO launched the web-based DHIS2 in 2006, based on open-source frameworks. Two technological shifts enabled the subsequent global uptake of DHIS2. First, mobile internet coverage grew rapidly in low- and middle-income countries (Sanner et al., 2012). Second, cloud computing allowed a web-based approach to implementation (Manya et al., 2012). Technology advances aside, the key enablers of DHIS2 adoption were persistent capacity building and knowledge sharing, involving university and government collaborations (Braa et al., 2004). Today, more than 70 ministries of health worldwide use DHIS2 for public health

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management, with impacts on areas such as data quality and use (Braa et al., 2012; Byrne & Sæbø, 2022; Nisingizwe et al., 2014), overall system strengthening (Dehnavieh et al., 2019) and disease surveillance (Kinkade et al., 2022; Reynolds et al., 2022). However, the DHIS2 venture encountered growing pains (Roland et al., 2017). First, the growing technical complexity of DHIS2 presented barriers to participation in software development and customisation. Second, managing software development became challenging due to divergent and competing needs and requirements. These issues informed the transition to DHIS2 as a platform in terms of architecture, organisational form, and governance.

The DHIS2 platform consists of customisable core applications for data capture, analysis, visualisation, messaging, data sharing, metadata management, data quality assessment and system administration. Globally distributed HISP groups support the participating ministries of health in the customisation and implementation of DHIS2, aligned with local resources and priorities. These HISP groups are autonomous organisations that vary in size (ranging from 6 to more than 70 members), areas of specialisation, geographical orientation and language proficiencies. The HISP UiO and the HISP groups—the HISP network—share implementation experiences and training materials, develop DHIS2 applications and collaborate to support country needs. Since 2011, more than 10 000 participants have attended capacity-building DHIS2 Academies. Academy specialisations include data-driven decision-making, data quality, software customisation, database administration, security and custom DHIS2 application development. The HISP UiO manages the DHIS2 App Hub (like an app store) for sharing custom DHIS2 applications developed by ecosystem actors.

3.2 | Data collection

Our study draws on a combination of naturally generated historical data and primary data collected in real time (Langley, 1999; Van de Ven & Huber, 1990). Participant observation notes, meeting minutes, email exchanges, and notes from semi-structured interviews and ad hoc conversations, including verbatim quotes, constitute our primary sources of data. Table 1 details our primary data collection activities, conducted between 2015 and 2022, with reference to six empirical episodes. An episode refers to a sequence of events that stand apart from others. We describe our identification of empirical episodes in the data analysis Section 3.3. In Table 1, the right column outlines our secondary sources of data.

The first author has followed the scale-up of DHIS2 development and implementation since 2010. This engagement includes participant observation in workshops and meetings, discussions in-person, via video conferences or by email with DHIS2 developers, project managers, HISP group representatives and officials from ministries of health. He has conducted recurrent field visits to nine countries with DHIS2 implementations, primarily in Sub-Saharan Africa. The field visits have focused on IS strengthening and integration, university collaborations and capacity building.

The second author is not directly involved with the organisations and processes that constitute the HISP network. He has primarily contributed to *post-hoc* data analysis in terms of theorising the findings and questioning the empirical chains of evidence, assumptions and interpretations of the other three authors.

The third author worked with DHIS2 on public health interventions in Zambia between 2012 and 2017. Since 2017, he has worked on DHIS2 software development coordination with the HISP UiO. In the latter role, he has solicited software requirements for a set of DHIS2 core applications. His experiences from representing both the demand and the supply sides of DHIS2 software development represent a unique perspective on participation in the DHIS2 requirement prioritisation process.

The fourth author has been involved in DHIS2 implementation since 2005. He has participated in numerous DHIS2 development and implementation projects and contributed to the establishment of arenas where the scope and purpose of DHIS2 have been deliberated by academics, developers, implementers, policymakers, government officials, NGOs, international donors and users. He provides a longitudinal perspective on how the DHIS2 network

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TABLE 1 Primary and secondary sources of data.

Primary data (2015–2022)	Secondary data (2010–2021)
Episode 1: Adoption of a software platform architecture (2012–2017)	
• Notes and minutes from 7 regular and ad hoc meetings with DHIS2 core developers in Oslo • 16 pages of hand written notes from software developer workshops and phone conferences	• Documents, including internal reports, spreadsheets and memos, generated from day-to-day operational activities of HISP UiO developers and HISP group representatives • HISP UiO strategy documents (seven) from 2015 onwards • DHIS2 data model documentation and roadmap blueprints from 2010 onwards
Episode 2: Development of an HISP memorandum of understanding (2016–2017)	
• Notes from 7 meetings with HISP UiO and HISP group representatives regarding software requirements, implementation support and partnership contracts • Notes and minutes from a 2016 workshop with HISP groups and HISP UiO regarding HISP group collaboration dynamics	• Documentation from DHIS2 annual conferences, workshops and meetings from 2015 onwards (9 conferences with 18 sessions on HISP collaboration) • Documentation from correspondence among HISP groups, prospective HISP groups, consultancies and international organisations regarding contracts, coordination and DHIS2 quality control
Episode 3: Decentralisation of DHIS2 academies (2017–2019)	
• Participant observation notes from 12 decentralised DHIS2 Academies (five days) across 8 countries • Notes from 5 coordination meetings between DHIS2 Academy participants and local HISP group facilitators	• Naturally generated documentation from meetings and workshops since 2010, where capacity building around DHIS2 has been a key theme (> 150 meetings) • DHIS2 Academy materials and documentation regarding their decentralised delivery by HISP groups from 2016 to 2019
Episode 4: Devolution of DHIS2 requirement prioritisation (2017–2020)	
• Participant observation notes and minutes from one HISP UiO strategy meeting concerning HISP groups' influence on DHIS2 software development • 14 pages of hand written notes concerning handling of DHIS2 requirements from an HISP group workshop in Oslo on 2–6 December 2019	• DHIS2 user and developer mailing lists and issue tracking software entries • Software release notes and issue history since 2010 • Documentation from joint fieldwork between HISP UiO and HISP group representatives in Asia and Sub-Saharan Africa to solicit requirements • Documentation from HISP strategy meetings concerning handling of DHIS2 requirements • Minutes from development milestones, product development and requirement prioritisation meetings • (> 450 meetings)
Episode 5: Introduction of the DHIS2 design system (2018–2021)	
• 12 pages of hand written participant observation notes from a one-week field visit to Tanzania in May 2019 • Notes from HISP group workshops in June and December 2019 concerning DHIS2 app development • 19 structured interviews with HISP group representatives concerning custom DHIS2 applications • - Notes from interaction with DHIS2 core software developers regarding the DHIS2 Design System and maintenance of custom applications	• Documents regarding operational activities of HISP UiO developers and HISP group representatives • HISP UiO strategy documents (eight) from 2015 onwards • Minutes (> 50) and documentation from bi-weekly HISP strategy meetings (2017–2020)

(Continues)

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TABLE 1 (Continued)

Primary data (2015–2022)	Secondary data (2010–2021)
Episode 6: Cultivation of the DHIS2 online community (2019–2022)	
• Synthesis of responses to a DHIS2 online community survey with 62 anonymous respondents • Notes from eight ad hoc conversations with online community contributors • Notes from two semi-structured interviews with the DHIS2 online community coordinator • User satisfaction survey results regarding the design and use of the DHIS2 online community (39 respondents)	• Discussion threads on online portals (i.e., DHIS2 online community and Jira) • DHIS2 online community usage statistics

has faced growing pains and how different strategies and tactics have been employed to navigate emergent constraints.

The engaged insider role of three out of the four authors calls for reflexivity. Our stakes in the project and long affiliation with the processes we describe naturally shape how we sort and engage with our rich data and how we interpret them. We are neither able to transcend our subjectivity nor do we regard it as a constructive path (Pillow, 2003). Our close engagement has been key to our understanding of how and why governance in the DHIS2 ecosystem has been carried out in certain ways. We are well positioned to understand the longitudinal developments of the DHIS2 platform and the HISP network, although 'position alone is insufficient to realise the potential' (Alvesson, 2003, p. 178). Indeed, some avenues can be closed for us, and insider researchers may even be perceived as subversive or following political agendas (Brannick & Coghlan, 2007). Nonetheless, being four authors with different roles, backgrounds, and levels of affiliation with the DHIS2 venture has allowed us to contest one another's views (Malterud, 2001). As engaged insiders primarily on the supply side of the DHIS2 platform, we have sought to balance our perspectives with the voices of the demand side. In our data, these voices are represented by the accounts of HISP groups and other ecosystem actors, who, in turn, represent the concerns of their ministry of health partners.

3.3 | Data analysis

DHIS2 has evolved from a small participatory design project in one health district in South Africa in the 1990s to a multicontinental and multisector web-based digital platform today. At the outset of our inquiry, we were concerned with how DHIS2 was able to support value creation across increasingly heterogeneous actors and settings, while maintaining a shared code base, sense of purpose and direction. The path from this initial interest to the data collected and the analysis presented in this article can be reconstructed into three stages.

The first stage entailed an open analysis, whereby we drew on historical data and our experiences to produce data displays (Miles & Huberman, 1994). The displays sketched out roles, relationships, practices and coordination mechanisms in the HISP network at different intervals from 2010 onwards. We focused on how mediation between the design and use of DHIS2 had changed with an increasing scale, with HISP groups taking a prominent role. Through iterations between overarching data displays and deep dives into empirical examples, we identified three areas of activities that characterised the transition of DHIS2 and the HISP network into a digital platform ecosystem. These activities comprised (1) *opening the software architecture* to support local problem solving, (2) *capacity building and knowledge sharing* throughout the HISP network and (3) *facilitation of participation in software development and implementation*.

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The three areas of activities triggered connotations with seminal works on innovation network orchestration (e.g., Dhanaraj & Parkhe, 2006; Lusch & Nambisan, 2015; Nambisan & Sawhney, 2011). Specifically, Dhanaraj and Parkhe (2006) have outlined orchestration processes that afford value creation in networks where the coordinator lacks the 'authority to issue commands and autonomous network members are not obliged to obey' (p. 660). We were inspired by this work, which emphasises governance processes over structure, although the focus on economic value creation did not align with the empirical realities of our case. Our case featured extensive pro bono work on increasing resource-constrained actors' capacity to participate in value co-creation around DHIS2, as well as the unprompted sharing of innovation and knowledge among the actors in the ecosystem.

At the second stage of the analysis, we directed our attention to the relations among governance processes, governance mechanisms, platform evolution and value co-creation (de Reuver et al., 2018). It is important to highlight two aspects of the analysis at this stage. First, throughout the history of DHIS2, its proponents and participants have been driven by an explicit goal to ensure that local innovations are fed back into a pool of open and shared resources. This aim informed our exploration of the literature on commons governance, particularly Frischmann's (2012) work on the relation between social options and social value. Frischmann's ideas allowed us to approach enabling social options as a guiding principle informing governance processes in the DHIS2 platform ecosystem.

Second, to synthesise years of empirical insights, we employed a temporal bracketing approach (Langley, 1999). We drew inspiration from Newman and Robey's (1992) process model that views development as a series of episodes characterised by triggering events and responses. We used the idea of a triggering event as an analytical tool to identify and demarcate episodes where existing governance mechanisms constrained the realisation of social options or obscured innovation processes. These limitations were observed in the form of unmet local needs, the lack of meaningful participation or misalignment between collective values and the activities of individual actors. To qualify triggering events, we considered the severity of the constraint and the level of engagement by ecosystem actors to prompt a governance response.

A triggering event marks the beginning of an episode and signifies a governance opportunity to address the experienced constraint. For example, the introduction of a DHIS2 Design System (Episode 5) was informed by the realisation that the open APIs and their documentation were insufficient to enable local development and maintenance of DHIS2 applications. Through our temporal analysis, we selected six episodes of pronounced constraints and governance responses, allowing us to delineate their distinct characteristics. Taken together, the episodes illustrate both the particularity (Episodes 1–4) and complementarity of the governance processes (Episodes 5 and 6).

At the third stage of the analysis, we focused on emerging patterns across the six episodes. This led us to conceptualise three governance processes—related to the three areas of activities identified at the first stage of the analysis—as *resourcing*, *capacitating* and *purposing*. *Resourcing* signifies an opening and broadening of the digital platform's scope and diversity, which is consistent with Ghazawneh and Henfridsson's (2013) original coining of the term. *Capacitating* allows ecosystem actors to participate in the identification and realisation of social options, while *purposing* seeks to align individual actors' activities and outputs with communal values. Based on the literature that emphasises the processual character of platform governance (e.g., Eaton et al., 2015; Leong et al., 2024), we have observed that governance mechanisms are created and modified over time to mitigate emergent constraints to the *ecosystem-wide enablement* of social options. We thus identify resourcing (**R**), capacitating (**C**) and purposing (**P**) as distinct *processes that shape* governance mechanisms, which, in turn, affect the ecosystem actors' activities and outputs towards the enablement of social options. Table 2 summarises the six empirical episodes and identifies their triggering events, changes in governance mechanisms, and the new structures and patterns of activities that enable social options to be innovated and realised.

4 | FINDINGS

In this section, we present six episodes concerned with the governance of the DHIS2 platform ecosystem. The episodes characterise resourcing, capacitating and purposing processes. The platform leader instigates these processes

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TABLE 2 Analysis of six episodes.

Episode	Triggering event	Changes in governance mechanisms	Enabling social options
Episode 1: Adoption of a software platform architecture (2012–2017)	The entire DHIS2 development is carried out by HISP UiO, which is regarded as a serious bottleneck.	**R**: HISP UiO provides open APIs, SDKs, an App Hub and tutorials	DHIS2 can be extended through applications, enabling the deferral of design and use decisions.
Episode 2: Development of an HISP Memorandum of Understanding (2016–2017)	Rapid DHIS2 uptake challenges responsiveness to local needs in a fair and transparent manner.	**P**: An MoU is developed to ensure alignment across HISP groups and uphold the HISP ethos.	The MoU explicates roles and responsibilities in the HISP network, endorsing reciprocity, sharing and social development.
Episode 3: Decentralisation of DHIS2 Academies (2017–2019)	HISP UiO is unable to cater to the demand for advanced DHIS2 capacity building.	**C**: Decentralisation of DHIS2 Academies grants wider participation, while localisation of content ensures relevance.	HISP groups customise DHIS2 Academy materials locally and contribute to a shared repository of generic materials.
Episode 4: Devolution of DHIS2 requirement prioritisation (2017–2020)	Country representatives lament the lack of transparency and responsiveness with core DHIS2 development.	**P**: Structures for representation of country requirements are put in place across product streams and domains.	HISP groups prioritise and propagate country requirements, which sensitise the DHIS2 roadmap to country needs.
Episode 5: Introduction of the DHIS2 Design System (2018–2021)	Boundary resources fall short in supporting development and maintenance of custom applications.	**R, C and P**: The DHIS2 Design System is developed and coupled with advanced training on application development.	Custom application development grants actors autonomy to act locally, while sharing and maintaining common features globally.
Episode 6: Cultivation of the DHIS2 online community (2019–2022)	Existing resources are inadequate in meeting the demand for implementation support.	**C and P**: The DHIS2 online community is open for anyone to join and share needs, experiences and opinions.	Experienced implementers provide guidance and support, while software features and ideas can be vetted with the community.

when social options are not being realised in local contexts or when local value creation fails to feed back to the shared pool of platform resources.

4.1 | Episode 1: Adoption of a software platform architecture (2012–2017)

For the sake of simplicity, Episode 1 hones in on one group of software features, commonly referred to as the DHIS2 Dashboard. These features allow system end users' side-by-side analysis of programmatic indicators. The DHIS2 Dashboard was originally conceived through an interaction between district health officers in Kerala state in India and HISP India developers in 2006. The module was initially regarded as 'a hack' by HISP UiO developers. However, the features were deemed useful and re-engineered into DHIS2 releases from 2009 onwards. Despite its broad uptake and numerous requests for its improvement, the DHIS2 Dashboard underwent scant revisions in the following years. A small team of HISP UiO developers wrote all of the code for official releases, which had become a serious bottleneck. This prompted work on open APIs, which were launched in 2012.

In late 2016, HISP Tanzania launched an Interactive Dashboard application that presented a radically different dashboard experience. The application addressed the requirements advocated by the Ministry of Health in Tanzania,

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including the ability to toggle between different visualisation types and dimensions. HISP Tanzania presented the application at workshops and shared it through the DHIS2 App Hub. Reflecting on the development experience, a HISP Tanzania representative noted difficulties with the DHIS2 version support, complicated development frameworks and limitations with the APIs. Furthermore, he argued, '*Our main challenge is the dire need for innovation funding. We often run into conflict with our timesheet-based project management*'.

While HISP groups developed custom applications in response to local needs, few innovations were shared with the DHIS2 community. Sharing an application on the App Hub was likely to trigger requirements from other projects, without earmarked funding. Whereas HISP UiO development received funding from a consortium of international donors, the HISP groups relied on implementation project grants. A representative from HISP Vietnam noted the challenge of supporting custom applications: '*We need to think through how it would work when country "A" uses an app developed for country 'B'. Who pays for features, maintenance and bug fixes?*'

In 2017, the HISP UiO redesigned the bundled DHIS2 Dashboard application and drew inspiration from the Tanzania Interactive Dashboard. This feature appropriation resolved, to some extent, the noted issue of local application maintenance. The innovation's originators and everyone else gained access to a stable application, with HISP UiO performance updates and support. As noted by a senior product manager at the HISP UiO, '*An innovation happens in the community, and we look at it and drag it into [the platform] core [development] if it's viable for everyone. It's a safe place to be because the original application developers have already worked out the use cases [and] user experience and vetted the functionality*'.

4.1.1 | Analytical summary

Episode 1 describes how *resourcing* opened DHIS2 development. An open API allowed distributed responses to local needs and highlighted new features that could be incorporated into the digital platform. At the same time, the episode shows limitations with resourcing since the scarcity of capacity and funding, as well as cooperation challenges, restricted the distributed maintenance of custom applications. Essentially, despite the open APIs affording local solutions to project requests and needs, the HISP UiO largely continued to present a bottleneck in terms of feature appropriation, limiting the volume of innovations that could be availed of as open, generic and shared platform resources.

4.2 | Episode 2: Development of an HISP memorandum of understanding (2016–2017)

Since its inception, the HISP has been organised as an informal network. The rapid scale of DHIS2 uptake and the need to enter contractual agreements with ministries of health led to the establishment of legal entities called HISP groups,[3] often springing from concrete implementation activities. Alumni from master's and doctoral programmes at the HISP UiO typically established HISP groups. Episode 2 concerns the formalisation of an HISP Memorandum of Understanding (HISP MoU) in 2017, which aimed to regulate cooperation in the HISP network.

With the growing demand for technical support, entities without ties to the HISP network started to offer training and consultancies, sometimes claiming to be endorsed by the HISP UiO. This situation muddied the waters for those seeking DHIS2 expertise. International agencies, which were financially supporting country implementations, requested the HISP UiO to vouch for trusted entities, such as through certification. At the same time, the HISP UiO desired to protect the HISP ethos and to promote DHIS2 experts who shared the same values and concerns. Additionally, in 2016, HISP groups expressed the need to formalise ground rules for closer cooperation. An HISP group

[3]At the time of writing, there are 20 recognised HISP groups.

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representative commented, *'There seems to be an absence of a unifying 'glue' between the groups—[there is] neither an ideology nor [a] shared principle or anything else—hence, each group is an island'*.

This state of affairs informed the development of an HISP MoU.[4] In the document, signatories are described as '*long term and trusted partners located in developing countries and collaborating with [the HISP UiO]*'. The HISP MoU consists of 19 points, which can be grouped into three sets of principles or values, important for enabling social options. The first principle states, 'Give priority to activities strengthening sustainable national health information systems'. This principle is followed by this pledge: *'Focus on the use of information to improve health service delivery*. These principles articulate *what* DHIS2 *should* be used for. The second set of values focuses on participation by fostering local entrepreneurship, supporting researcher access and engaging users in design and development. The third set of values is concerned with reciprocity and sharing in the form of open-source software, generic applications, training materials and knowledge.

The MoU promotes social options through sharing. What is produced by signatories should not only be provided as public goods but should also be generic to allow their appropriation across contexts. HISP groups are incentivised to sign the MoU. Implementation projects, which constitute the main source of funding, become accessible to trusted signatories. Signatories benefit from their visibility at webinars and conferences, access to career development through degree programmes at collaborating universities and influence on the DHIS2 development roadmap. Consultancies that leverage DHIS2 but operate outside the framework of the MoU are free to develop proprietary DHIS2 applications and services that do not comply with norms of reciprocity, sharing and local ownership.

4.2.1 | Analytical summary

The HISP MoU was developed as a collaborative effort between the HISP UiO and HISP groups to explicate previously taken-for-granted norms and values in the expanding ecosystem. The MoU is a governance mechanism instigated through a *purposing* process. It serves as both a commitment to cooperation in the HISP network and a way to differentiate trusted HISP groups from actors with unknown capacities and motives. It thus signifies alignment across a limited number of ecosystem actors around a set of core values, while keeping other actors at arm's length.

4.3 | Episode 3: Decentralisation of DHIS2 academies (2017–2019)

DHIS2 Academies cover advanced DHIS2 implementation and customisation topics. They are organised on-site with participants from whole regions, such as East Africa, West Africa, and Latin America. The HISP UiO had organised and conducted all DHIS2 Academies since their inception in 2010. However, by 2016, the number of planned academies for that year reached 17 without quenching the demand. This additional burden on the HISP UiO informed the transfer of its responsibility for DHIS2 Academy delivery to the HISP groups.

In consultation with HISP groups, the HISP UiO developed standard curricula and facilitation guidelines for DHIS2 Academies across thematic areas. The number of DHIS2 Academies swelled to 27 in 2017. Throughout the following year (2018), HISP groups reported discrepancies between the standardised curricula and participants' needs. With the DHIS2 Academies organised into narrow topics, the participants who rarely attended more than one academy did not gain the insights required to engage in value co-creation around DHIS2. For example, training sessions on advanced DHIS2 Analytics were ineffective since the participants would return to their respective countries to work on datasets and databases to which those analytics did not apply, and they had limited awareness of how to mitigate the experienced gaps.

[4]Available at https://www.mn.uio.no/hisp/english/networks/.

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> *We feel that we didn't link the system's design to the actual output. In covering the analytic tools, we did not get into the foregoing design. When you are designing, you are designing for the output. Should we combine the design and customisation academy with the analytics academy?*
>
> (HISP West and Central Africa representative)

HISP groups modified some course materials to cater to interdependencies across topics and to include locally relevant content, databases and user stories. For example, HISP Uganda and HISP Tanzania introduced materials related to locally developed custom applications, such as the Interactive Dashboard mentioned in Episode 1. In parallel, HISP groups worked with the HISP UiO to incorporate local materials into the global curriculum. The DHIS2 Academies thus mirrored the development of the digital platform itself. The HISP UiO maintained core content, while HISP groups developed and shared local enhancements. Despite the decentralised delivery of the DHIS2 Academies and HISP groups' day-to-day support to ministries, some countries' reports indicated that capacity-building offerings were insufficient. To mitigate these inadequacies, The HISP UiO developed and launched self-paced, free, online Academies for basic DHIS2 training as well as a few advanced topics.

A number of independent initiatives deliver DHIS2 training programmes, without any ties to the HISP network. However, some initiatives struggle to gain traction or maintain engagement. For example, despite being one of the earliest DHIS2 training providers, starting in 2012, Akros Inc. eventually suspended its DHIS2 training activities as the DHIS2 Academies became more streamlined.

> *We stopped training to go in a different direction. We really enjoyed the early days of DHIS2 because there was great room to explore use cases, present results and opportunities, find different best practices and the like. As the global body of knowledge grew, there was less to 'explore', and the focus [of the venture] shifted to incremental change, knowledge maintenance, etc.*
>
> (Akros representative)

There are lingering concerns that the HISP UiO is too involved in decisions about which DHIS2 Academies to provide, as well as when and where to do so, and who should deliver and participate in them. This centralised coordination creates a situation where actors that might be better positioned to respond to growing capacity-building demand in specific areas—but are outside the HISP purview—fail to be recognised as legitimate or motivated to contribute.

4.3.1 | Analytical summary

A normative focus on *whom* to grant participation in the realisation of social options around DHIS2 has motivated the HISP UiO to perform a coordinating and gatekeeping role in relation to advanced capacity building. The *capacitating* process reported in this episode led to the decentralisation of DHIS2 Academies to ensure their responsiveness to local needs and to grant countries with limited resources fair access to training opportunities. Coordination from the HISP UiO contributes to the perception that DHIS2 Academies are of high quality and grants participants a closer relationship with the platform leader, thus partly undermining the establishment of alternative training programmes.

4.4 | Episode 4: Devolution of DHIS2 requirement prioritisation (2017–2020)

At the DHIS2 Annual Conference in Oslo in 2015, with participants from ministries of health, international NGOs, and HISP groups, a session was held on core development and how to influence it. A lead developer presented six

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criteria used by the HISP UiO to weigh DHIS2 requirements: perceived usefulness, estimated workload, expected improvement over existing features, perceived complexity, expected level of participation from the requestor and funding availability. Based on these criteria, it is not surprising that resourceful late adopters of DHIS2 had become influential in shaping the DHIS2 roadmap. International NGOs, which encountered DHIS2 through engagement with their ministerial partners, started to adopt it for their internal reporting and data management needs. The same NGOs invested in face-to-face visits with core developers in Oslo to ensure that their needs and requirements were well understood. A UiO researcher, who has been involved from the early days of DHIS implementation in South Africa, commented, *'Writing DHIS2 requirements is not easy; it has to be a dialogue'*.

By 2017, HISP groups lamented that requirements from less resourceful countries were not sufficiently addressed. An HISP group representative remarked, *'For ministries, we are the interface of the DHIS2 community and, as such, we feel the heat and embarrassment over bugs and features promised and not delivered'*. This situation informed a *purposing* response. An HISP UiO reflection document described the need to *'assure that country requirements are considered for the DHIS2 roadmap' and that 'representatives from each HISP group will offer advice on priorities'*. An advisory team of HISP group representatives was formed to participate in virtual requirement handling meetings. However, after a series of deliberations, version 2.28 of DHIS2 was released in October 2017, with minimal implementation of the team's recommendations. While a process had been put in place to articulate DHIS2 requirements, its effects were meagre. Furthermore, it did not bring much transparency to how the HISP UiO made final roadmap decisions. The virtual meetings were discontinued because the complexity of the software and the heterogeneity of country needs made prioritisation untenable. One HISP UiO implementation coordinator reflected on the situation:

> [The HISP UiO] has no one in the field! No one is looking closely—what are the gaps? We need people to look for requirements within the ministries and then work closely with the developers. We need to narrow the gap between the user and the developer or specifically between the HISP groups and the core development teams.

In December 2019, during a one-week workshop with HISP groups in Oslo, the participants discussed how to exercise a stronger influence on the software development process. The representative from HISP Rwanda asked, *'How can I signal the critical importance of my request?'* The representative from HISP Uganda added, *'System needs are growing; the requirements from the field are not prioritised. The requirements we prepare, when we then look at the plans, these things are nowhere [...]; small things you feel should be fixed quickly, take time, and then, the project is over'*.

In response, the HISP UiO established an annual roadmap development process. In September each year, HISP groups coordinate with their ministerial partners to produce a ranked list of feature requests across DHIS2 product streams. The HISP UiO provides a template for describing and ranking requests. Requirement refinement sessions are conducted to assist in the articulation of requirements. Each HISP group then presents its list to DHIS2 product managers. Each request is given a score that factors in the rank of the request. The product managers then convene to consider technical feasibility, coordinate any cross-product dependencies and ultimately produce an integrated roadmap. The final roadmap addresses the highest-scoring requests, which are channelled into milestones and bi-annual releases. Finally, the roadmap is published on the DHIS2 website, and a webinar is conducted to relay it to the community.

4.4.1 | Analytical summary

A modularised and stepwise approach to requirement handling contributed to roadmap transparency. The *purposing* process gave weight to the needs of ministries with limited capacities and resources in shaping the development of the DHIS2 platform, while moderating the influence of resourceful actors. Transparency in the requirement-handling

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process allowed HISP groups to better manage implementation projects expectations and engage in effective realisation of social options around the DHIS2 platform.

4.5 | Episode 5: Introduction of the DHIS2 design system (2018–2021)

In late 2018, to strengthen support for custom application development, the core development team started working on a design system, with build tools and libraries of common user interface (UI) components, such as buttons, progress bars and data entry fields. These resources were expected to improve usability, stability and security; ease application development; and facilitate consistency. In early 2019, HISP Tanzania began working on yet another dashboard application in response to local project requirements. The Bottleneck Analysis Dashboard utilised the design system. *'Half of the application is based on reusing core components; this saves us time to focus solely on the new functionality. The application looks and behaves more like the core Dashboard application that everyone already knows'* (HISP Tanzania representative).

However, several HISP groups reported that the UI library and design system guidelines were difficult to use. A representative from HISP West and Central Africa argued for the need to *'lower the entry barrier to those wanting to build apps'*. Furthermore, custom application maintenance remained time consuming. For HISP India, the maintenance of custom applications posed a serious challenge. *'Everything is project based. Requirements just keep coming. Our requirements are very time bound compared to the core team. We must do custom applications. There are [custom] applications for anything not supported in the core'* (HISP India representative).

These concerns informed a combination of resourcing and capacitating responses. In 2019, the DHIS2 core team extended the design system by introducing its App Platform. This core team maintains the App Platform components and libraries. Changes made on the App Platform are mirrored in custom applications. Hence, the App Platform transfers a portion of the application maintenance work to the DHIS2 core team.

Up to this point, custom application development had primarily been carried out based on individual developers' personal preferences. The DHIS2 Design System introduced common procedures and frameworks that contrasted with such idiosyncratic ways of development. In December 2019, during an HISP group workshop held in Oslo, a session was dedicated to custom application development. The HISP UiO Application Development Lead invited a discussion by saying, *'If there is friction between developers in the field and developers building the [DHIS2 Design System], which should be making things easier, there is a need to talk'*. An implementation coordinator from the HISP UiO suggested, *'We must invest time from the core side to work with local teams. Not only through training, but also mentorship. Ensuring that [core] developers spend time on training and capacity building is difficult because it takes time away from [the] development of features'*. A representative from HISP Uganda responded, *'It would be great if developers in Uganda, who are trying to develop according to DHIS2 standards, can be more integrated into the development process'*.

The discussion highlighted a key challenge. With its limited development capacity, the HISP UiO must strike a balance between developing the platform core, refining the DHIS2 Design System, and providing mentorship in custom application development. Based on requests from HISP groups, the core team opted to emphasise mentorship to utilise the DHIS2 Design System, which resulted in a series of DHIS2 Developer Academies and workshops. Following the establishment of the DHIS2 Design System, the number of applications shared on the DHIS2 App Hub swelled to 75. Interestingly, all of the 17 custom application developers interviewed in 2021 indicated that the platform's sharing ethos was a key motivation for developing generic applications and making them publicly available through the DHIS2 App Hub, as illustrated by the following quotes:

> *We have to think beyond our own interests and benefits, but also that of the community.*
>
> (Independent software company representative)

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We like that others can benefit from our solutions.

(HISP Sri Lanka representative)

We've been inspired to be generic from the global DHIS2 philosophy. The whole philosophy of HISP, most of us started from a software development where we worked directly with the core developers in Oslo. We've been part of this design thinking. It comes with the whole history. It is implicit.

(HISP Tanzania representative)

However, the maintenance challenges described in Episode 1 still lingered. Out of the 17 interviewees, all but one indicated that the organisations they represented had no financial motives to produce a generic DHIS2 application, and they expressed their desire to hand over their applications to the HISP UiO. The following two quotes illustrate these views:

The financial motivation [to develop generic applications] is not there at all. When [an] HISP group pushes an application to the App Hub, we are obliged to support the application, and when anyone needs any help, we help them out. We'll get funding for building the application but not the maintenance and support.

(HISP Vietnam representative)

If the core absorbs something, then it is a success. The only gap is how we can be recognised [for absorption].

(Independent software company representative)

Long-term financial sustainability of custom applications has emerged as a limiting aspect of the DHIS2 ecosystem. The HISP UiO has limited capacity to absorb the maintenance of applications. As a partial remedy, pooling resources among ecosystem actors has become more commonplace. For example, an application that embeds training materials directly in DHIS2 UIs was jointly developed by international NGOs and subsequently shared on the DHIS2 App Hub. Similarly, some HISP groups have agreed to co-develop applications from a shared roadmap. For example, HISP Uganda, HISP Tanzania and HISP Rwanda together maintain the Bottleneck Analysis Dashboard application mentioned earlier. However, no global solution has been identified, and some actors see their custom applications become deprecated and discarded due to a lack of earmarked funding.

4.5.1 | Analytical summary

Episode 5 describes a *resourcing* process that attempted to both lower the entry barrier to application development and streamline application maintenance. A *capacitating* effort was introduced in parallel in the form of Developer Academies and mentoring. Finally, the governance mechanisms were shaped by *purposing* to encourage actors to develop custom applications in a generic manner and afford appropriation by others. In combination, the resourcing, capacitating and purposing processes shaped the governance mechanisms of the platform in such a way that custom application development became partly disentangled from application maintenance. This process lowered the barrier to application development, which made it possible to realise social options around the platform and reduced the cost and burden of sharing a custom application with other ecosystem actors.

4.6 | Episode 6: Cultivation of the DHIS2 online community (2019–2022)

Despite the nascent pooling of resources, as shown in Episodes 3 and 5, cross-fertilisation and knowledge sharing among country implementations were limited and often relied on facilitation by the HISP UiO in the form of

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workshops or other events. Since the early days of DHIS2 development, the HISP UiO has maintained an online resource for the coordination of developers' work and for tracking software issues. However, the technical orientation of this resource excluded actors with a strong focus on implementation. Towards the end of 2018, the HISP UiO launched the DHIS2 online community to allow DHIS2 implementation knowledge to be co-created and shared in a distributed and ad hoc manner. The HISP UiO provides the online hosting, infrastructure and a few facilitators to the community's more than 3000 members. An online community member with some implementation experience recounts:

> Mostly, I get help with system customisation. There is not a lot of flexibility in what we're doing in the field, so I appreciate reading the stories and other use cases, but more frequently, I just need to know how to make a particular thing work.

The DHIS2 online community members build relations across organisations and projects. HISP groups present their areas of expertise and join forces to solve implementation challenges. However, after 1 year, two shortcomings of the online community were discernible. First, few members responded to the more demanding issues and requests. As one online community member noted, *'The frontline online community managers are super supportive, but I rarely find solutions when the topic is more complex'*. Consequently, some community members expressed the need for more advanced support and training, as illustrated by this comment: *'We still need more engagement from UiO implementers and developers. Sometimes, calls between members when needed, more user experience demos and more sessions with core developers'* (South Sudan Ministry of Health representative).

Second, there was no direct link between the online community and Jira, the issue-tracking tool used by software developers. One member distinguished the online community from Jira by referring to the former as *'a discourse or commentary rather than diagnosis and feature request'*.

In response, three adjustments were made. First, community facilitators were tasked with translating community issues into Jira tickets. The community facilitators maintained communication with developers and product managers to ensure that both these groups' responses were relayed back to the community. Second, a virtual arena was developed to share the roadmap in extensive detail. The same arena allowed online community members to propose and vote for features on the DHIS2 roadmap. The community vote is now weighed together with the HISP group priorities (Episode 4), technical feasibility and other factors to inform roadmap prioritisation. Finally, community facilitators organise monthly virtual meetups where community members discuss pressing concerns and share practices. Reflecting on the changes, a representative from an NGO in Zambia noted, *'The best thing about joining the community is learning and troubleshooting issues from a diverse group of users during the monthly meetups. I now also have the privilege of sharing and voting for new features in DHIS2'*.

Online subcommunities are often formed in the wake of topical DHIS2 Academies (Episode 3), such as Disease Surveillance, Malaria, HIV, Logistics and Education. While this setup provides community members with virtual spaces to congregate, the HISP UiO employs purposing through the centralised establishment of new discussion topics, thus limiting the scope of deliberation to those areas that are deemed important to the community. Through this normative shaping, new members become exposed to the ideological roots of the HISP network and how DHIS2 is currently employed, in what domains and for what purposes.

4.6.1 | Analytical summary

The DHIS2 online community provides an arena for ad hoc sharing of knowledge and experiences among a growing number of distributed heterogeneous actors, who typically have limited resources and opportunities to convene on physical locations. Episode 6 illustrates how *purposing* and *capacitating* can be carried out in tandem and shape governance mechanisms to both afford participation in value co-creation activities and align those activities with

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communal goals and values. The online community serves as a platform for espousing values associated with the HISP network, thus influencing what is perceived as appropriate value realisations from the open and generic pool of digital platform resources.

The six episodes have touched on fundamental questions about the DHIS2 ecosystem. What is DHIS2? What should it be used for? For and by whom is it being developed? The answers to these questions are not settled once and for all. Rather, they are continuously contested as ecosystem actors encounter constraints in their efforts to realise and innovate social options. For social options to be realised, the value propositions around DHIS2 need to be translated into meaningful local actions across user organisations, application areas and domains. The episodes have highlighted interdependencies among the ability to extend DHIS2 locally, community-wide knowledge sharing, fair handling of software requirements and continuous deliberations about what DHIS2 is and should become. In the following discussion, we relate our findings to the governance dilemma of balancing the autonomy of heterogeneous actors to create social value locally with collective action towards communal goals.

5 | DISCUSSION

To frame our inquiry, we asked, *How can governance mechanisms be shaped to enable social options in a digital platform ecosystem?* Investigating this question, we identified three processes—*resourcing*, *capacitating* and *purposing*—through which governance mechanisms were established and altered over time to enable social options. Based on our findings, we claim that *governing platform ecosystems for social options involves continuous shaping of governance mechanisms to mitigate emergent constraints to such options' realisation and innovation*. To delve deeper into what such governance entails, we first elaborate on enabling social options as a guiding principle for platform governance. We then discuss the three identified governance processes in relation to existing literature. Finally, this allows us to outline a model that shows the role that these processes play in governing for social options.

5.1 | Enabling social options as a guiding principle

Governing for social options entails deferring design and use decisions to actors at the fringes of the ecosystem (Frischmann, 2012). The rationale is that platform leaders do not – and cannot – have the requisite knowledge to produce social value across heterogeneous settings but can support and favour those who aspire to do so. A key implication of our inquiry is thus to establish the *enablement of social options* as a guiding principle for governance in digital platform ecosystems geared towards the creation of social value.

Digital platform ecosystems provide a technological foundation and an organisational form that enables realisation and innovation of social options on a global scale, across resource asymmetries and divides. The common premise of the digital platform literature is that platforms enable distributed value co-creation in an ecosystem of heterogeneous actors (e.g., de Reuver et al., 2018; Eaton et al., 2015; Tiwana, 2013). The perspective on social options extends this premise in three ways. *First*, the value produced by realising social options intentionally exceeds individual interests and benefits a wider community (Chamakiotis et al., 2021). Our findings illustrate this. For example, in Episode 2, groups in the HISP network signed an MoU and committed themselves to strengthening national health ISs and promoting the use of information to facilitate fair public service delivery. This pledge follows the HISP network's aspiration to support countrywide implementation coverage and to ensure that governments with scarce resources have an equal opportunity to inform the development trajectory of the shared digital platform.

Second, enabling social options hinges on cooperation among distributed actors. Cooperation has been identified as a key governance concern in knowledge-intensive digital platform ecosystems, such as enterprise software (Foerderer et al., 2019; Huber et al., 2017; Sarker et al., 2012; Wareham et al., 2014). Cooperation involves balancing actors' self-interests against common goals (Huber et al., 2017; Saadatmand et al., 2019; Wareham et al., 2014). In a

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digital platform ecosystem geared towards the generation of social value, cooperation is essential to enable downstream innovation of social options. Examples of actors' close cooperation to harness innovation are cited in Episodes 1, 3 and 5, where HISP groups and other ecosystem actors share, build on and maintain one another's work (e.g., custom applications and training materials) to foster downstream identification and realisation of social options.

Third, governing for social options presents novel value co-creation dynamics between platform leaders and contributing actors. Prior studies of digital platforms have identified the concerns involved in a platform leader's decision to appropriate individual actors' innovations to maximise an ecosystem's value (Schreieck et al., 2021; Wareham et al., 2014), at the risk of driving contributors away (Ceccagnoli et al., 2012; Nambisan et al., 2018; Parker & Van Alstyne, 2017; Schreieck et al., 2021). Our findings contrast these observed dynamics. HISP groups encouraged the platform leader's appropriation of local innovations. Appropriation was perceived as beneficial to both the wider community and the originator since it afforded long-term maintenance and support of desirable features. For actors in resource-constrained settings, this opportunity provided an incentive for developing generic solutions to local problems and for sharing their work. Actors could focus on local implementation needs, with fewer concerns about maintenance, which increased the social options available to them.

To summarise, enabling social options as a guiding principle for platform governance steers the ecosystem towards collective action in social value creation that favours a community's needs over individual interests. In the following two subsections, we consider how shaping governance mechanisms entails balancing between ecosystem actors' autonomy to respond to local needs and their cooperation in collective enablement of social options.

5.2 | Governance processes: Resourcing, capacitating and purposing

We identify three distinct processes through which governance mechanisms are shaped to enable social options. *Resourcing* denotes the creation and modification of predominantly structural boundary resources in the form of layered modular architectures, standardised interfaces and automation (Ghazawneh & Henfridsson, 2013; Gregory et al., 2018) that open up the platform for local experimentation and tinkering. Resourcing is a prerequisite for local appropriation and value co-creation with any digital innovation platform (Dittrich, 2014; Roland et al., 2017; Wareham et al., 2014) and provides flexibility for developing heterogeneous complements (Eaton et al., 2015; Ghazawneh & Henfridsson, 2013).

Prior research on complex software platforms has identified the need to manage intricate value co-creation dynamics (Parker et al., 2016; Sarker et al., 2012; Wareham et al., 2014). These studies have shown that simply granting access to the digital platform, through structural boundary resources, is insufficient to enable value-adding complements (Huber et al., 2017). Rather, value emanates from the interplay between local expertise and knowledge-intensive platform resources (Bergvall-Kåreborn & Howcroft, 2014; Foerderer et al., 2019). In our case, the establishment of open APIs (Episode 1) and the development of the DHIS2 Design System and the App Platform (Episode 5) lowered the entry barrier to innovation and afforded custom applications to be made generic, consistent and maintainable. These initiatives allowed distributed actors to focus their efforts on the local appropriation of shared platform resources rather than bargain for software feature updates. However, resourcing alone did not produce significant improvements in terms of realisation and innovation of social options in the DHIS2 ecosystem.

With *capacitating*, we observe a similar pattern of distributing resources and decisions among ecosystem actors to enable social options. The DHIS2 Academies represent a devolution of DHIS2 capacity building to HISP groups. Ecosystem actors produce and share new training materials and cooperate in the delivery of DHIS2 Academies. The DHIS2 Academies also contribute to fostering shared norms and values that promote decentralised public health decision-making and support health service delivery. Similarly, the DHIS2 online community lowers the bar for accessing and contributing to the DHIS2 knowledge base. It provides an arena for implementation problem solving, experimenting with ideas (Benkler, 2013; Frischmann, 2012) and expressing opinions on DHIS2 development.

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Furthermore, the online community provides a place for actors with different areas of expertise to make themselves visible and seek collaboration opportunities, peer approval and recognition.

Capacitating entails shaping governance mechanisms to enable participation in the identification and realisation of social options. It involves favouring communities with limited resources and pronounced needs over resourceful actors with individualistic agendas. This position contrasts with governance in digital platform ecosystems geared towards the creation of economic value, where partnership programmes, workshops, and direct support (Bergvall-Kåreborn & Howcroft, 2014; Foerderer et al., 2019; Huber et al., 2017; Sarker et al., 2012), are offered to actors most likely to boost network effects and profits (Schreieck et al., 2021; Wareham et al., 2014). When governing for social options, knowledge-intensive boundary resources are primarily focused on lowering the entry barriers and facilitating fair and meaningful participation in social value creation.

Directing the realisation of social options towards social value is further pursued through *purposing*. The establishment of a HISP MoU exemplifies this process since it directs signatories towards strengthening primary healthcare in collaboration with ministry of health partners. The MoU directly promotes social welfare and serves as soft gatekeeping vis-à-vis actors who do not share the same visions and values. Furthermore, MoU signatories commit to sharing what they produce and cooperating towards DHIS2 development and implementation, fostering generic resources for downstream social value generation. This commitment thus complements efforts to open the digital platform through resourcing and overcome barriers to participation through capacitating.

Whereas previous research has reported how enterprise software platform leaders introduce partner hierarchies based on each actor's economic value co-creation potential and level of identification with the ecosystem (Wareham et al., 2014), we identify a different form of prioritisation of actors based on their alignment with the communal values of social development, sharing and reciprocity. This favouritism promotes enabling social options within the DHIS2 community, while sacrificing the generative potential of actors and agendas that are not closely aligned with communal values. The devolution of the DHIS2 roadmap prioritisation process (Episode 4) demonstrates how influence on the software development trajectory is aligned with communal values by giving weight and priority to otherwise underrepresented country needs and by assisting actors with limited resources in the articulation of requirements.

The purposing processes in our case focus on communal values. The HISP UiO leverages social capital and relational control to cultivate a cooperative atmosphere. In other words, it seeks to endorse *collective action* (Ostrom, 1990). Purposing informs the normative development of content, training materials and documentation. Value-laden experience sharing and socialisation opportunities are available through DHIS2 Academies, webinars, mentorship, the DHIS2 online community and an annual DHIS2 conference (Episodes 3 and 6). Relational control in a digital platform ecosystem is made possible by the pre-existence of shared values (Tiwana, 2013). In the case of the HISP network, these values are rooted in a public sector ethos and an orientation towards social development. Most actors who seek to contribute to the DHIS2 community already work in resource-constrained settings to strengthen public services. A shared identity is further harnessed through investments in personal development, such as training and mentoring opportunities and master's and doctoral programmes.

We have identified three governance processes and how they shape governance mechanisms to enable social options. As presented in the findings (Section 4), we have observed how governance in the DHIS2 ecosystem responded to constraints in the enablement of social options. We now turn to the temporal dynamics of governing for social options.

5.3 | A process model of governing for social options

The evolution of governance mechanisms in digital platform ecosystems is not driven solely by the unilateral actions of platform leaders (Eaton et al., 2015; Karhu et al., 2018). Platform leaders need to assert control over the ecosystem, while accommodating innovative behaviour (Tiwana, 2013). The heterogeneous actors in a digital platform

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ecosystem pursue their individual goals of meeting the changing and diverse needs of end users. In doing so, actors encounter constraints, which may inform governance responses that shape the governance mechanisms of the ecosystem (Eaton et al., 2015; Leong et al., 2024; van der Geest & van Angeren, 2023). The platform leader has direct influence on governance mechanisms, while other actors can turn to collective or political action to seek accommodation for their goals (Eaton et al., 2015). Our findings concur with this evolutionary view of ecosystem governance. However, our study highlights the governance processes that unfold differently from digital platforms primarily geared towards the generation of economic value. To account for this distinction, we derive a process model of governing for social options (Figure 2).

Our model emphasises how governance mechanisms in platform ecosystems geared towards enabling social options evolve to overcome the constraints experienced by heterogeneous actors in the realisation and innovation of social options. The episodes in our study show this dynamic and the need for new and modified governance mechanisms to maintain a feedback loop between local value creation and the enablement of social options. Constraints to the enablement of social options thus *informs* a governance response involving one or more of the three governance processes. For example, Episode 3 illustrates how the failure to meet the demand for advanced DHIS2 capacity building constrained the realisation of social options. This situation informed a capacitating process in the form of simultaneous standardisation and decentralisation of DHIS2 Academies to the HISP groups. The pattern emerging from our findings is that the identification of unmet community needs, due to the lack of available routes to realise social options, informs resourcing. Constrained participation—in the sense that the actors comprising the ecosystem are unable to realise social value—informs capacitating. Misalignment *between* individual actors' activities and outputs *and* the ecosystem's communal values informs purposing.

Our model provides insights into how the identified governance dilemma of balancing local autonomy with collective action in a digital platform ecosystem can be navigated. This dilemma is never resolved but requires continuous governance where new mechanisms are created and existing ones are modified. As with other collective action problems, this involves a combination of different governance approaches over time (Markus et al., 2006; Monge et al., 1998). While actors in the DHIS2 ecosystem pursue individual goals, resourcing, capacitating and purposing are governance processes through which individual goal-seeking is balanced with guiding principle of enabling social options. A platform architecture with standardised interfaces enables contributing actors to make local platform enhancements, which are important preconditions for local value realisation (Eaton et al., 2015; Ghazawneh & Henfridsson, 2013). However, the availability of structural boundary resources is insufficient for ensuring that local

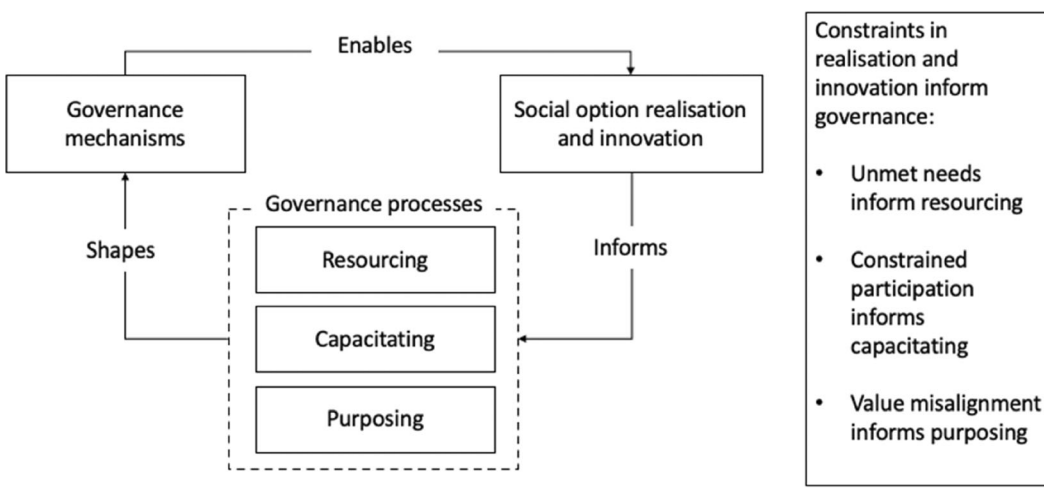

FIGURE 2 A process model of governing for social options.

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value creation enable further social options. In the case of DHIS2, this insight has informed governance mechanisms where the platform leader takes over the maintenance of popular applications and an MoU that functions as soft gatekeeping for implementation and consultancy contracts. Investing in actors' capabilities through training opportunities, while offering design systems to lower the entry barriers, raises ecosystem actors' capacity to participate in the realisation and innovation of social options. In a community-oriented digital platform ecosystem, such as DHIS2, the heterogeneous actors that comprise its ecosystem continuously negotiate the collective understanding of its purpose and the scope of its value creation potential.

5.4 | Contributions to research

Our first contribution is an alternative perspective on governing value co-creation in digital platforms where the potential to cooperatively develop and share common resources to solve local problems is at the core. What such problems are and how they are tackled will vary across contexts, as will the understanding of what comprises social value. We have employed and further developed Frischmann's (2012), p. 49) concept of social options to describe what the platform leader, striving for the creation of social value, must achieve through platform governance. Social options constitute enabling conditions for contextually appropriate social value creation and depend on the digital platform's characteristics, the ecosystem of actors contributing to it, and locally available resources. We believe that the concept of social options is valuable for further research on increasingly vast and interconnected digital ecosystems governed by one or a few focal actors. A social options perspective is perhaps particularly relevant to the study of governance towards collective action vis-à-vis digital public goods (Nicholson et al., 2022).

Our second contribution to research lies in addressing the paucity of empirical studies of digital platforms with a social welfare or social development agenda (Bonina et al., 2021; Chamakiotis et al., 2021). A common aspect of such digital platforms is that they are geared towards the creation of social value (Chamakiotis et al., 2021). For policymakers, our study provides an empirical example of how an open digital platform architecture, a platform organisation, and a governance regime that enables social options constitute a digital platform ecosystem intended to create social value. We believe that extrapolations from our case can inform 'platform thinking' in the public or non-profit sectors (Brown et al., 2017; Cordella & Paletti, 2019; Gong & Li, 2023; Thompson & Venters, 2021). Our case highlights the need for platform leaders in ecosystems geared towards the creation of social value to collaborate closely with ecosystem actors, respond to their experienced constraints and ensure that governance mechanisms align actors' activities and outputs with communal values.

Our third contribution lies in showing through our process model how governing for social options requires continuous attention. Our analysis suggests that balancing the activities of individual actors with collective action can be achieved through governance processes that respond to constraints to the community's realisation and innovation of social options. Specifically, we have illustrated how unmet needs inform *resourcing*, constraints in participation inform *capacitating* and value misalignment informs *purposing*. This model extends platform research that emphasises the processual character of platform governance (e.g., Eaton et al., 2015; Fürstenau et al., 2019) and opens opportunities for further studies into how digital platform ecosystems geared towards the creation of different forms of value require various forms of governance over time.

The fourth contribution of our study is derived from our insider perspective and the empirical insights this affords. There is rich literature on digital platforms (Constantinides et al., 2018; de Reuver et al., 2018), where researchers have utilised deep empirical engagements with platform leaders (e.g., Schreieck et al., 2021) and innovative research methods (e.g., Eaton et al., 2015) to build insights into how platform ecosystems are governed. However, there are few research accounts by engaged insiders on the supply side of digital platforms, and none that we are aware of that focuses on social value creation. Therefore, our findings contribute rich and novel empirical insights into platform governance through a longitudinal, uniquely situated and engaged insider account (Brannick & Coghlan, 2007).

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6 | CONCLUSION

We have empirically accounted for three governance processes in a digital platform ecosystem where open, generic and shared resources are developed for health management purposes across more than 70 low- and middle-income countries. Our study highlights the platform leader's role in recognising constraints to the realisation and innovation of social options and in shaping governance mechanisms to help actors overcome such barriers. Governance mechanisms that enable social options must align efforts to open and broaden the scope of the digital platform to accommodate emergent needs, capacitate ecosystem actors to participate in identifying and realising social options, as well as foster cooperation in generating shared knowledge and innovation. Essentially, governance mechanisms should enable both distributed autonomy and facilitate collective action over time. Governing for social options is thus a continuous balancing act.

Our study has limitations that need to be considered. It has been carried out by insiders on the supply side of a digital platform, which necessarily has had implications for both data collection and analysis. As insiders, we possess unique insights into the longitudinal evolution of the digital platform. However, our roles also potentially hinder us from obtaining in-depth knowledge about alternative perspectives in the ecosystem. Being based on a single case, the findings are also characterised by the HISP network's specific history and context. DHIS2 is a platform that is primarily used for health management purposes in the public sector, and the platform leader (HISP UiO) is a centre at a university. The platform's context and direction towards social forms of value make our analysis of governing for social options most relevant in relation to community-oriented platforms orchestrated by NGOs and public sector actors. However, engaging with existing literature that takes a process perspective on platform governance (e.g., Eaton et al., 2015; Fürstenau et al., 2019) leads us to propose that our process model is also relevant to a broader set of digital platforms where value co-creation is dynamic and requires cooperation among ecosystem actors to address complex issues over time.

We suggest that further research can be carried out on the proposed process model in other digital platform ecosystems oriented towards social value creation to learn whether similar dynamics can be identified over time. Another limitation of our study is that we have not investigated patterns of the interplay among the three governance processes. Future research could therefore examine potential patterns of this interplay and investigate if there are discernible sequences over time, involving resourcing, purposing and capacitating processes when governing for social options in digital platform ecosystems.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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